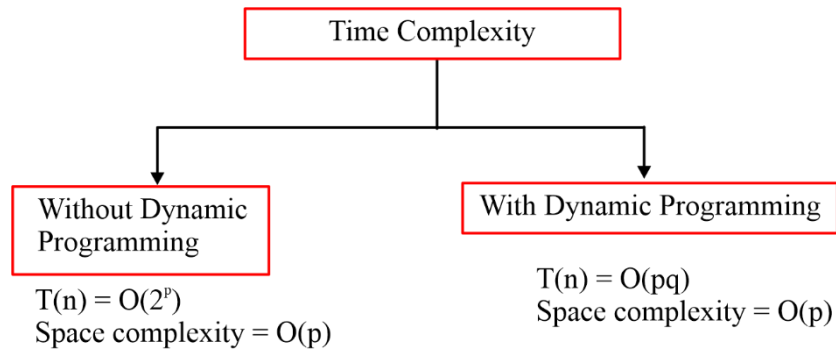


4.8 0/1 Knapsack Problem

The maximum profit can be achieved by 0/1 knapsack problem where capacity of problem is 'p' and number of objects are 'q'.



4.9 Sum of Subset Problem

- Given n-elements and an integer 'm', it is required to determine whether there exists a subset of given n elements, whose sum equal M.
- This is a decision problem (True/False).

4.9.1 Algorithm for Sum of Subset Problem

SoS(n, M, A)

// A [1 . . . n] is an array of elements

```

{
1. for i = 0 to n
    for j = 0 to M
        if (i >= 0 and j = 0)
            SoS [i, j] = T
        else
            if (i = 0 and j > 0)
                SoS [i, j] = F;
            else
                if (A[i] > j)
                    SoS [i, j] = SoS [i - 1, j]
                else
                    SoS [i, j] = SoS [i - 1, j] or
                    SoS [i - 1, j - A[i]]
            }
}
  
```



4.9.2 Time Complexity of SoS

Two for loops are there thus repeating for $(n * m)$ times. Thus, time complexity = $O(n * m)$

Time complexity of SoS becomes exponential if $M = 2^n$

$$T.C = O(n \times 2^n)$$

